BY ORDER OF THE COMMANDER OF GRAND FORKS AIR FORCE BASE (ACC)

GRAND FORKS AIR FORCE BASE INSTRUCTION

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Operations

RESPONDING TO SEVERE WEATHER INSTRUCTIONS

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This instruction implements guidance for Responding to Severe Weather Events. This instruction describes the 319th Reconnaissance Wing procedures. It applies to individuals at all levels who prepare, manage, review, certify, approve, disseminate and/or use official Air Force publications and forms, including all personnel assigned, attached or associated with the 319th Reconnaissance Wing. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR), 319th Reconnaissance Wing Plans and Programs (319 RW/XP), 701 Eielson St., Grand Forks Air Force Base, ND 58205, using the AF IMT 847, Recommendation for Change of Publication; route AF IMT 847s from the field through the base publications/forms manager (319 CS/SCXK). Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with AFMAN 33-363, Management of Records, and are disposed of in accordance with the AFRIMS Records Disposition Schedule (RDS) at **https://www.my.af.mil/afrims/afrims/afrims/rims.cfm**. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document is substantially revised and must be completely reviewed.

This instruction has been changed to reflect changed roles and responsibilities of Air Force Weather (AFW) organizations "Mandatory" severe weather criteria have been changed to "Standard" severe weather criteria to allow commanders flexibility in determining actions taken as owners of the Operational Risk Management process.

This document is substantially revised and must be reviewed in its entirety. It presents detailed guidance and procedures to follow prior to severe weather events. AFI 90-201, The Air Force Inspection System, 2 August 2013 no longer uses the term Exercise Evaluation Team (EET). This instruction replaces EET with Wing Inspection Team (WIT).

Chapter 1—INTRODUCTION **4**

1.1. General. 4

1.2. General Information. 4

Chapter 2—RESPONSIBILITIES **5**

2.1. The Installation Commander: 5

2.2. All supported on and off base agencies: 6

Table 2.1. Travel Condition Chart 6

2.3. Supporting Wing Inspection Team (WIT): 9

2.4. Supporting Civil Engineer Readiness Unit: 9

2.5. Supporting Communications Unit: 9

2.6. The Supporting OWS and CWT Together: 9

2.7. 319 RW Staff Agency: 10

2.8. 319 RW/CP: 10

2.9. 319 RW/PA: 11

2.10. 319 RW/HC 11

Table 2.2. 319 RW/HC (Chapel) Services And Programs 11

2.11. 319 MDG/CC: 12

2.12. 69RG/CC: 12

2.13. 69 MXS/CC: 12

2.14. 319 MSG/CC or 319 MSG/CD: 12

2.15. 319 FSS/CC: 13
2.16. 319 LRS/CC:..... 13
2.17. 319 SFS/CC: 13
2.18. 319 CONF/CC: 14
Chapter 3—WEATHER WATCHES AND WARNINGS 15
3.1. Criteria. Table 3.1 15
Table 3.1. Standard Watch and Warning Criteria..... 15
3.2. Minimum Desired Lead-time..... 16
3.3. Weather Watch. 16
3.4. Weather Warning..... 16
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 17
Attachment 2—SEVERE WEATHER RESPONSE PROCESS 19
Attachment 3—MATRIX OF WEATHER CONDITIONS REQUIRING WATCHES OR WARNINGS 20
Attachment 4—QUICK LOOK TEMPERATURE REACTION MATRIX 21
Attachment 5—WIND CHILL TEMPERATURE (WCT) INDEX CHART 23
Attachment 6—MINIMUM EXTREME COLD WEATHER PROTECTIVE ISSUE 24
Attachment 7—EXTREME COLD ILLNESS PREVENTION 25
Attachment 8—USE OF NEW GENERATION HEATERS IN WARMING SHELTERS 27
Attachment 9—WINTER STORMS AND EXTREME COLD SAFETY 28
Attachment 10—LIGHTNING SAFETY 34
Attachment 11—HIGH WIND SAFETY 35
Attachment 12—LARGE HAIL SAFETY 36
Attachment 13—CHEMICAL BIOLOGICAL WARFARE DEFENSE (CBWD) COLD WEATHER OPERATIONS 37
Attachment 14—COLD WEATHER FACILITY INSPECTION PROCEDURES 38

Chapter 1

INTRODUCTION

1.1. General. Each year Air Force installations, equipment, and personnel are threatened by severe weather. When severe weather occurs, it can affect readiness and the ability to accomplish Air Force missions. Severe weather is defined as any weather phenomena requiring a special severe weather notice from an Air Force Weather unit. The severe weather notice is designed to give the designated agency advance notification (with sufficient time to implement protective actions) of expected weather conditions that pose a hazard to personnel, resources, and property. When notified of the potential for severe weather, the agency will normally take subsequent preventative actions. **Table 3.1** contains standard watch and warning criteria.

1.2. General Information. Successful preparation and response to severe weather events require an integrated approach among operations, logistics, and support units on a base. The key aspects to ensuring units optimally protect people and equipment lie in ensuring all base elements fully understand their role in the prediction, notification, preparation, or response to severe weather. Units must identify how they will respond to severe weather that affects the installation and geographically separated operating locations and ensure these plans are exercised. Key actions to consider in protecting people and equipment are: identifying mission limiting/damaging weather criteria; forecasting specific severe weather events by issuing watches and warnings; disseminating the watches and warnings in a timely manner; and taking required protective actions based on the watches and warnings (See **Attachment 2**). Finally, by documenting and cross-feeding lessons learned, severe weather resource protection is enhanced.

Chapter 2

RESPONSIBILITIES

2.1. The Installation Commander:

2.1.1. Will chair meetings as required, but not less than annually, to review installation severe weather preparedness, capabilities, requirements, and procedures. Meetings will be attended by representatives of base agencies responsible for developing protective action plans, CWT, guard and reserve components, tenant organizations, local off-base units, and others organizations with severe weather notification requirements. Other attendees will include installation agencies responsible for disseminating the information (e.g., base communication squadron and CP). Meetings will address at a minimum the following issues:

2.1.1.1. Severe weather observing and forecasting capabilities and corresponding user requirements to identify the weather phenomena for which notification is required to include threshold values and desired lead-times.

2.1.1.2. Primary and back-up dissemination procedures.

2.1.1.3. Protective action procedures and resources.

2.1.1.4. Severe weather awareness training and exercise procedures.

2.1.2. Will ensure installation personnel are educated on the local severe weather threat and applicable protective measures, as well as on the purpose, applicability, and operating procedures of the watch-warning system. The training will ensure personnel authorized to issue “back to work” orders following a stop work order are properly equipped to assess weather hazards and appropriate precautions. Educational information can be included in newcomer orientation briefings, base newspaper articles, commander’s call briefings, the base web site, and other forums deemed appropriate by the installation commander.

2.1.3. Will ensure base agencies develop and document pre-severe weather protective measures and post-severe weather response plans. Requirements and support will be documented and recertified annually for each phenomenon in weather support documents, wing instructions, host-tenant support agreements, or similar documents (e.g., base annexes). These plans/agreements will include a base warning notification pyramid structure diagram. The notification pyramid structure will include all on and off-base agencies (e.g., swimming pools, golf courses, child care centers) requiring support.

2.1.4. Will ensure CWT leadership is available to assist in the severe weather management process by prioritizing assignment of additional and temporary duties not related to weather (i.e., security forces augmentees, squadron safety manager, etc.).

2.1.4.1. Will ensure sufficient CWT personnel are available to provide “eyes forward” support to the OWS during severe weather episodes. The CWT will provide back-up weather watch and warning support to their installation on behalf of the OWS IAW AFMAN 15-129. Providing “eyes forward” support and base resource protection has a higher priority of duty than assignment to base details and additional duties out of the weather facility.

2.1.5. Will ensure the appropriate alternate weather observing site is available with necessary work- space to meet customer specified required notification timelines.

2.1.6. Will ensure OWSs and/or CWTs are required to telephonically contact as few agencies as possible to relay severe weather information. The use of a pyramid-type reporting system should be employed whenever possible and maximum use of technology is encouraged (i.e., the use of local area networks, e-mail) to facilitate the relay of information.

2.1.7. Will conduct periodic exercises tailored to upcoming seasonal weather or other environmental concerns in order to evaluate the timeliness of personnel notification and response capability for both on and off-base agencies and geographically separated units. Include a test of the primary and back-up dissemination systems not tested as a result of real world events. Integration of these exercises with base/post disaster preparedness response exercises is encouraged.

2.1.8. Will report severe weather events at the installation IAW OPREP-3 reporting procedures in AFI 10-206.

2.1.9. Will forward base-wide after-action reports to their MAJCOM/FOA/DRU following severe weather events and exercises if lessons learned offer valuable cross-feed information which highlights:

2.1.9.1. Quality and timeliness of the forecast.

2.1.9.2. Dissemination and notification successes and/or shortfalls.

2.1.9.3. Base-wide resource protection response (e.g., protection checklists).

2.1.9.4. Applicable lessons learned.

2.1.10. Will report problems or concerns with the technical capabilities of the OWS, and/or CWT. If appropriate, request MAJCOM assistance in assessing their technical capabilities.

2.1.11. The Installation Commander or Vice Commander will determine when Airfield and Air Traffic Control Operations will cease due to severe weather based off the recommendation of the 319 OSS/CC.

2.2. All supported on and off base agencies:

2.2.1. Will develop procedures to notify all activities and personnel of impending severe weather events and ensure understanding of appropriate weather terminology.

2.2.2. Will create and document pre-severe weather protective measures, as well as post severe weather response plans, for their operations. As a minimum, procedures developed will address:

2.2.2.1. “Travel Condition” chart identifies Travel Condition Green, Yellow and Red. **Table 2.1** applies to all 319 RW personnel (Coordination via MDG/BE)

Table 2.1. Travel Condition Chart

Travel Condition	GREEN	YELLOW	RED
Visibility	Unrestricted	50 ft to ½ mile	<50 ft

Snow, Ice, Mud, Hail, Water on Road, or Severe Weather	None or Negligible	Hazardous but Passable	Almost Impassable
Travel Actions	Unrestricted	Use Caution	Travel Actions

2.2.2.2. Actions to safeguard personnel, including severe weather shelter locations.

2.2.2.2.1. Will implement procedures to seek shelter from severe weather to safeguard personnel and locations when required. The following are considered severe weather conditions: 50KT winds, freezing precipitation, blizzards, extremely cold wind chills (-48 and colder), and thunderstorms that produce tornadoes. Thunderstorms that produce hail with a diameter of 3/4" and greater is also considered severe weather criteria. Use **Attachments 2** through **14** to help develop plans.

2.2.2.3. Will take actions to safeguard aircraft, equipment, and other supplies stored outdoors.

2.2.2.3.1. Actions for safeguarding and securing aircraft, equipment, and other supplies stored outdoors, as required to minimize damage to the resource as well as nearby facilities.

2.2.2.4. Will take actions to protect facilities.

2.2.2.4.1. A listing of all facilities managed, actions for protection of facilities, and procedures for conducting building checks in accordance with **Attachment 14**.

2.2.2.5. Will take actions to take prior to resuming operations following a severe weather episode.

2.2.2.6. Will provide a listing of key activities as defined in **paragraph 2.4.5.1**

2.2.2.7. Will provide a listing of all organization non-key work and training activities that can be cancelled, when necessary, due to the severity of weather.

2.2.3. Will provide initial/refresher severe weather training to all personnel every year. Personnel will be educated on the local severe weather threat, applicable protective measures, and operating procedures of the watch-warning system. Use **attachments 2** through **14** as training material. Document all training on individual's AF Form 55, Employee Safety and Health Record, or other Air Force approved system.

2.2.4. Will budget for, and issue, at minimum, the items listed in **Attachment 6**. All personnel must keep these items readily available from September through March to safely accomplish outdoor work, outdoor training requirements (e.g. CATM training), and for personal protection during emergencies (e.g. car accidents, emergency evacuation of buildings).

2.2.5. Will provide a listing of mission and storm essential activities to their respective group commander no later than September each year for group commander review. Commanders/directors will identify the non-mission essential activities that are not required to remain open during severe weather. Closing non-key activities is at the discretion of the commander/director.

2.2.5.1. Mission and storm essential activities are the activities identified by commanders/ directors that must continue to provide mission support, logistics support, medical support, operational support or emergency/contingency operational support during severe weather. Stopping these activities due to severe weather would severely degrade the base's ability to provide essential services or support. The commander/director has assessed the risks and determined that mission accomplishment takes priority over the increased personal safety of workers by requiring them to remain on duty during severe weather conditions.

2.2.6. Will identify and notify mission and storm essential military and civilian personnel who must report for duty/remain on duty to perform mission essential activities during severe weather. Provide a listing of identified civilian employees to 319 FSS/CPO no later than September each year. Commanders/directors are required to notify civilian employees of their mission, storm or cold weather designations and will also notify those personnel who assigned mission and storm essential activities when they are not required to report for duty/remain on duty during severe weather.

2.2.7. Will implement procedures listed in **Attachment 7** to prevent illness due to extreme cold temperatures.

2.2.8. When organizational owned activities effect external base customers and are not identified in this instruction, will recommend Courses of Action (COA) to respective group commander when severe weather watches or warnings are received.

2.2.9. Will enforce the buddy system for outside workers as indicated in **Attachment 7**.

2.2.10. Will notify all personnel under their command of severe weather watches, warnings, and return to work notices.

2.2.11. Will cancel all organization non-mission and storm essential work and training activities when appropriate.

2.2.12. Will ensure military personnel remain on telephone stand-by through end of normally scheduled duty hours if released early due to severe weather.

2.2.13. Will ensure civilian employees are approved administrative leave through end of normally scheduled duty hours if released early due to severe weather or the entire duty day if notified not to report for duty because of severe weather. Storm essential personnel are not normally released with charge to annual leave unless additional approval is granted in coordination with the Squadron/Group/Wing Commander

2.2.14. Will ensure government vehicle operators and passengers have necessary adverse weather clothing and survival kits in possession if leaving Grand Forks AFB during actual or forecasted winter storm or extreme cold weather.

2.2.15. Will issue back to work orders after completing an operational risk assessment indicating it is safe to do so.

2.2.16. Will provide 319 OSS/OSW a listing of weather phenomena for which notification is required, to include threshold values and desired lead times, no later than July of each year.

2.3. Supporting Wing Inspection Team (WIT):

2.3.1. Will assist the installation commander by developing and executing inspection scenarios. OWS participation in installation inspections is highly desirable, depending on workload priority.

2.4. Supporting Civil Engineer Readiness Unit:

2.4.1. Will provide guidance and advice to the WIT Chief .

2.4.2. Will ensure CEX assists the installation commander and 319 OSS/OSW personnel in thoroughly educating base agencies on the purpose, applicability, and operating procedures of the warning and watch system.

2.4.3. Will send reminder e-mails to Grand Forks AFB personnel for organizations to conduct facility checks and on-base housing residents to conduct home inspections as listed in **Attachment 14**.

2.4.4. Will recommend to 319 MSG/CC, when deemed appropriate, to suspend snow removal operations during severe weather for COA determination.

2.5. Supporting Communications Unit:

2.5.1. Will assist the local installation commander and CWT by ensuring primary and back-up communication systems are available to facilitate timely dissemination of weather information to on-base and supported off-base customers/agencies.

2.5.2. Will maintain 24 hour communication line/network repair capability during severe weather. This is a mission and storm essential activity.

2.5.3. Will recommend to 319 MSG/CC or 319 MSG/CD, when deemed appropriate, to initiate Centralized Base Information Transfer Center (BITC) mail distribution for their COA development.

2.6. The Supporting OWS and CWT Together:

2.6.1. Will coordinate requirements for and ensure the timely issuance of weather watches and warnings with supported customers IAW AFMAN 15-129.

2.6.2. Will establish procedures to manage severe weather threats, to include recalling of weather personnel as required IAW AFMAN 15-129.

2.6.3. Will perform formal reviews of severe weather events IAW AFMAN 15-129. The OWS is OPR for the formal review; the relevant CWT will provide inputs and coordination.

2.6.4. Will provide the information listed below to the CP in order to prepare OPREP-3 reports as required for the unit commander.

2.6.4.1. The actual severe weather conditions experienced.

2.6.4.2. The forecast valid at the time of the occurrence to include any watches or warnings issued.

2.6.4.3. The operational status of meteorological and communication equipment (e.g., radar, wind sensors, NIPRNET, etc.) at the time of the event.

2.6.5. Will conduct and document periodic severe weather refresher seminars for weather personnel who cover upcoming seasonal forecasting and observation techniques, expected threats, and response procedures.

2.6.6. Will assist the installation commander and disaster preparedness personnel in thoroughly educating base/post agencies on the purpose, applicability, and operating procedures of the weather watch and warning system.

2.6.7. Will request assistance through their MAJCOM/FOA/DRU on severe weather forecasting problems not resolvable at the local level or for severe weather forecasting seminars, as required.

2.6.8. Will document times for periodic testing of the primary and back-up dissemination system by sending messages and documenting their receipt by key agencies.

2.6.9. Will provide subject matter experts in support of periodic tests and exercises in **paragraph 2.5.1**

2.7. 319 RW Staff Agency:

2.7.1. Before a recommendation is made to the 319 RW/CC on a severe weather event, the 319 MSG/CC or 319 MSG/CD will gather inputs from all group commanders on appropriate COA if event requires action.

2.7.2. 319 RW Staff Agencies. Refer to **Attachment 5**, Wind Chill Temperature (WCT) Index Chart for warming cycles in severe cold weather. Command decisions for continuance of outside activities, mission essential activities, and mission critical activities will be made based off of WCT guidelines.

2.8. 319 RW/CP:

2.8.1. Will ensure timely notification of severe weather watches and warnings:

2.8.1.1. 319 RW/CC or CV and Group CCs will be notified of all watches/ warnings via email (telephonically if seriousness warrants immediate notification, senior duty controller will make determination).

2.8.1.2. CP will pass Wind Chill Temperature (WCT) and cold stress condition (as necessary) via AtHoc (Primary) or e-mail Grand Forks Distro (Secondary) as appropriate.

2.8.2. Primary notification of weather watches/advisories/warnings from 319 OSS/OSW will be telephonic.

2.8.3. Will disseminate severe weather watches, advisories and warnings via AtHoc (Primary).

2.8.3.1. Will make individual notifications to tenant units that do not fall under any Group, Wing Staff Agency, listed on Grand Forks Distro (e.g. DHS).

2.8.3.2. Will make telephonic notifications to 24 hour agencies (e.g., BDOC, Fire Department, Airfield Management, etc.).

2.8.3.3. CP will also make base-wide notifications of severe weather via AtHoc Network Mass Notification System. AtHoc is the primary network notification system. Notifications via giant voice are at the discretion of the 319 RW/CC (or designated

alternate). Exceptions include pre-approved use of giant voice (e.g. Tornado Warning, Lightning within 5NM, etc).

2.8.4. When directed by the 319 MSG/CC (or designated alternate) CP will pass on the appropriate closure or release message via Snow Line, AtHoc and base AM Radio station. They will also notify PA to contact local television and radio stations with appropriate message.

2.8.5. At the 319 RW/CC's (or designated alternate's) discretion, CP will begin a pyramid alert IAW GFAFBI 36-3803, *Personnel Recall and Accountability Procedures*. No other time.

2.8.6. Will submit OPREP 3 reports as identified in **paragraph 2.8.4**

2.8.6.1. IAW **paragraph 2.3.8** of this instruction.

2.8.6.2. IAW AFI 10-206, Operational Reporting and all supplements.

2.8.6.3.) At 319 RW/CC or designated alternate discretion.

2.8.7. For severe winter weather events, will run the Snow Line, Travel Condition Change and Alert AM Radio checklists as appropriate. Contact CP to review snow line checklist and travel condition checklist.

2.9. 319 RW/PA:

2.9.1. For severe winter weather, when contacted by the 319 RW/CP, will use all electronic means available to update the base populace on reporting procedures and road conditions.

2.10. 319 RW/HC

2.10.1. When a severe weather watch or warning is received from CP, the Wing Chaplain will implement the appropriate COA for 319 RW/HC. In the absence of the Wing Chaplain, the NCOIC, Chapel Operations or Acting Wing Chaplain will implement the COA.

2.10.2. The below matrix applies to all 319 RW/HC (Chapel) services and programs.

Table 2.2. 319 RW/HC (Chapel) Services And Programs

WCT	Restrictions	Waiver Authority
-30º F and below	Cancel all chapel youth programs to include Sunday School and Confraternity of Christian Doctrine Studies	Wing Chaplain
-40º F and below	Cancel all chapel programs, except for Ash Wednesday, Holy Week or Christmas programs. NOTE: Worship Services will continue as long as the base remains open.	Wing Chaplain

2.11. 319 MDG/CC:

2.11.1. Before a recommendation is made to the 319 RW/CC on a severe weather event, the 319 MSG/CC or 319 MSG/CD will gather inputs from all group commanders on appropriate COA if event requires action.

2.11.2. 319 MDG units. Refer to **Attachment 5**, Wind Chill Temperature (WCT) Index Chart for warming cycles in severe cold weather. Command decisions for continuance of outside activities, mission essential activities, and mission critical activities will be made based off of WCT guidelines.

2.11.3. When severe weather warnings indicate WCT of -48º F or colder, the 319 MDG/CC:

2.11.3.1. Will adjust clinic appointments as required/requested to ensure patient/staff safety.

2.12. 69RG/CC:

2.12.1. Before a recommendation is made to the 319 RW/CC on a severe weather event, the 319 MSG/CC or 319 MSG/CD will gather inputs from all group commanders on appropriate COA if event requires action.

2.12.2. **Paragraph 2.14** applies to all 69 RG units.

2.12.3. At WCT -40º F or colder, the 69 RG/CC:

2.12.3.1. Will allow airborne aircraft to continue with planned mission but aircraft will return with enough fuel to hold on the ground and expect slower than normal aircraft recovery operations.

2.12.4. The 69RG will operate aircraft in severe weather in accordance with their guiding instructions and Technical Orders (TO).

2.12.5. 69 RG ground activities. Refer to **Attachment 5**, Wind Chill Temperature (WCT) Index Chart for warming cycles in severe cold weather. Command decisions for continuance of outside activities, mission essential activities, and mission critical activities will be made based off of WCT guidelines.

2.13. 69 MXS/CC:

2.13.1. 69 RG Aircraft Maintenance Units. Refer to **Attachment 5**, Wind Chill Temperature (WCT) Index Chart for warming cycles in severe cold weather. Command decisions for continuance of outside activities, mission essential activities, and mission critical activities will be made based off of WCT guidelines.

2.14. 319 MSG/CC or 319 MSG/CD:

2.14.1. Before a recommendation is made to the 319 RW/CC on a severe weather event, the 319 MSG/CC or 319 MSG/CD will gather inputs from all group commanders on appropriate COA if event requires action. If a COA is required and approved by the 319 RW/CC, the 319 MSG/CC or 319 MSG/CD will notify CP of special instructions. CP will disseminate weather watch/warning with special instructions.

2.14.2. When necessary, coordinate with the Grand Forks Public School Grounds and maintenance personnel to determine if base schools will open.

2.14.3. 319 MSG units. Refer to **Attachment 5**, Wind Chill Temperature (WCT) Index Chart for warming cycles in severe cold weather. Command decisions for continuance of outside activities, mission essential activities, and mission critical activities will be made based off of WCT guidelines.

2.15. 319 FSS/CC:

2.15.1. Will ensure lodging (for those residing off base) and dining services are available for all personnel manning mission and storm essential activities during severe weather conditions. Lodging and dining facility are considered key activities.

2.15.2. Will ensure child care is available for all military and civilian personnel manning key activities during severe weather conditions. Child care is considered a mission and storm essential activity.

2.15.3. When data storage space is available, publicize all severe weather travel conditions, restrictions, and precautions on the marquee near the front gate.

2.15.4. Will recommend to the 319 MSG/CC or 319 MSG/CD, when deemed appropriate, to close any moral and welfare recreation, child care, or other FSS activity that will affect external customers for his COA development.

2.15.5. Will recommend to the 319 MSG/CC or 319 MSG/CD, when deemed appropriate to activate the Emergency Family Assistance Command Center (EFACC) for his COA development.

2.16. 319 LRS/CC:

2.16.1. Will ensure transportation is available for all military and civilian personnel manning mission essential activities during severe weather conditions. This is considered a mission essential activity.

2.16.2. Will maintain 24-hr capability to order/issue/deliver priority parts. This is considered a mission essential activity.

2.16.3. Will maintain 24-hr capability to provide aircraft and ground fuel. This is considered a key activity.

2.16.4. Will maintain a 24-hr capability to repair and recover emergency and other priority vehicles being operated to accomplish key activities. This is considered a mission and storm essential activity.

2.17. 319 SFS/CC:

2.17.1. Will maintain all posts and base security patrols. The SFS/CC must weigh security concerns before making a recommendation to pull base security patrols for inclement weather to the RW/CC for risk acceptance. This is considered a key activity.

2.17.2. SFS will respond to aid in mission critical personnel moving to their places of duty on base when coordinated through the Base Defense Operations Center, as part of their security patrols.

2.17.3. Will monitor local public highway routes and on-base road conditions during severe weather by contacting the Grand Forks County Sheriff's Department or other appropriate agency and the North Dakota Department of Transportation website. Report hazardous road

conditions to the 319 MSG/CC or 319MSG/CD for COA development. The SFS/CC will recommend on-base road conditions on the below chart.

2.17.4. Will recommend to the 319 MSG/CC, when deemed appropriate, to cancel CATM training for COA development.

2.18. 319 CONF/CC:

2.18.1. Will maintain vendor lists for local supply purchases to meet pre-severe weather requirements and post-severe weather recovery needs.

Chapter 3

WEATHER WATCHES AND WARNINGS

3.1. Criteria. Table 3.1 outlines the standard criteria which normally require weather watches and warnings. This list is not all inclusive and customers have the responsibility to make their individual requirements known at the local severe weather preparedness meeting or following the procedures outlined in AFMAN 15-129. Customer requirements must be identified in the WSD (or equivalent documentation). In addition, the CWT documents the requirements with the supporting OWS via a memorandum of agreement (or equivalent document).

Table 3.1. Standard Watch and Warning Criteria

STANDARD WEATHER WATCH CRITERIA	**STANDARD WEATHER WARNING CRITERIA**
Weather watches are issued when the following severe weather conditions are possible, but occurrence, location and/or timing is still uncertain. Watches are issued as a special notice that the potential exist s for the development of customer specified severe weather within the forecast area during the specified valid time. Watches normally precede a weather warning.	Weather warnings are issued when the following severe weather conditions exist, are imminent, or are highly likely to occur. Warnings are issued for customer specified operationally significant severe weather conditions for the forecast area which constitute a threat to personnel or property.
CRITERIA	**CRITERIA**
Potential for tornadoes	Tornadoes
Potential for high winds (≥ 50 knots, or locally established severe criteria).	High Winds (≥ 50 knots, or locally established severe criteria).
Hail (> 3/4 inch diameter, or locally established severe criteria).	Hail (> 3/4 inch diameter, or locally established severe criteria).
Potential for heavy Rain/Snow (> 2 inches in < 12 hours or locally established criteria).	Heavy Rain/Snow (> 2 inches in < 12 hours or locally established criteria).
Potential for Freezing Precipitation (any intensity).	Freezing Precipitation (any intensity).
Potential for Blizzard Conditions (Duration > 3 hours, sustained winds or gusts ≥ 30 knots, considerable falling and/o r b lo wing snow, with prevailing visibility frequently < 1/4 mile/0400 meters -all criteria must be met)	Blizzard Conditions (Duration > 3 hours, sustained winds or gusts ≥ 30 knots, considerable falling and/or blowing snow, with prevailing visibility frequently <1/4 mile/0400meters -all criteria must be met)
Potential for Lightning within 5 nautical miles (or locally established severe criteria).	Lightning within 5 nautical miles (or locally established severe criteria).
NOTE: Lighting watches and warnings are mandatory IAW AFOSH Standard 91-100.	

3.1.1. Non-mission/storm essential, mission/storm essential operations will be suspended or terminated at the discretion of the 319 RW/CC. Before a recommendation is made to the 319 RW/CC on a severe weather event, the 319 MSG/CC or 319 MSG/CD will gather inputs from all group commanders on appropriate COA if event requires action.

3.2. Minimum Desired Lead-time. Agencies require advance notice prior to the onset of a particular weather phenomenon. This is the minimum time required to take action to protect resources, property, and personnel. CWTs will coordinate and establish the actual values and timing for watches and warnings according to supported agency requirements. AFMAN 15-129 contains further information on standard watches and warnings, lead-times, and severe weather processes and procedures.

3.3. Weather Watch. This is special notice provided by an OWS or CWT to supported agencies alerting them to the potential for weather conditions of such intensity to pose a hazard to life or property. The weather watch can be thought of as a "heads up", at which time agencies need to consider implementing required protective actions should a subsequent weather warning be issued.

3.3.1. Upon issuance of a weather watch, all key personnel and agencies in high risk activities and operations should be notified of the potential of severe weather. Activities include, but are not limited to, agencies having aircraft, missiles, spacecraft, explosives, or petroleum; those performing open-air work, recreational activities, and underground utility work.

3.3.2. Upon receipt of a weather watch, operations may continue. However, personnel in high risk areas should be prepared to implement corresponding required protective actions without delay if and when a subsequent weather warning is issued.

3.4. Weather Warning. A weather warning is a special notice by an OWS or CWT to supported agencies giving them advance notification (with sufficient time for protective actions) of expected weather conditions of such intensity to pose a hazard to personnel, resources, and property.

3.4.1. Upon issuance of a weather warning, all key personnel and agencies in high risk activities and operations should be notified of predicted mission impacting or operationally significant weather conditions. Activities include, but are not limited to, those agencies having aircraft, missiles, spacecraft, explosives, or petroleum; open-air work, those performing recreational activities, and underground utility work.

3.4.2. Upon receipt of a weather warning, required protective actions corresponding to the weather event will be immediately implemented to safeguard personnel and equipment.

PAUL E. BAUMAN, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 10-206, *Operational Reporting*

AFMAN 15-129, *Aerospace Weather Operations-Processes and Procedures*

Prescribed Forms

None

Adopted Forms

None

Abbreviations and Acronyms

AFOSH—Air Force Occupational Safety and Health

AFW—Air Force Weather

BITC—Base Information Transfer Center

CATM—Combat Arms Training and Maintenance

CBRNE—Chemical Biological Radiological Nuclear and Explosive

COA—Course of Action

CBWD—Chemical Biological Warfare Defense

CWT—Combat Weather Team (installation weather Unit)

DFP—Defensive Fighting Positions

DRU—Direct Reporting Unit

EET—Exercise Evaluation Team

FOA—Field Operating Agency

FP—Force Protection

IAW—In Accordance With

MAJCOM—Major Command

MOA—Memorandum of Agreement

MOPP—Mission Oriented Protection Posture

NGH—New Generation Heater

OPR—Office of Primary Responsibility

OWS—Operational Weather Squadron

WSD—Weather Support Document

WF—Weather Flight

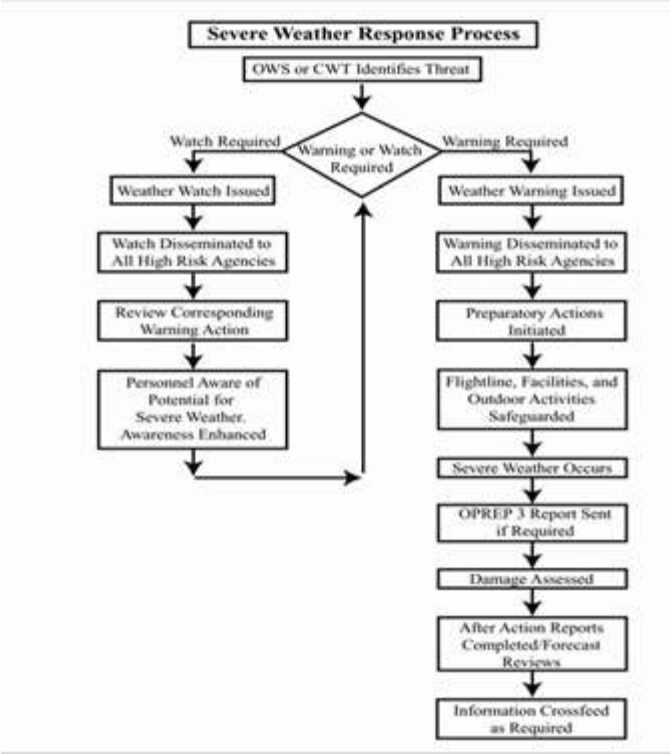
WCT—Wind Chill Temperature

WIT—Wing Inspection Team

Attachment 2

SEVERE WEATHER RESPONSE PROCESS

Table A2.1. Severe Weather Response Process



Attachment 3

MATRIX OF WEATHER CONDITIONS REQUIRING WATCHES OR WARNINGS

Table A3.1. Matrix of Weather Conditions Requiring Watches or Warnings

Weather Criteria	Watch Desired Lead-Time	Warning Desired Lead-Time
Tornado	As potential warrants	15 minutes
Winds > 50 kts	As potential warrants	60 minutes
Winds > 35 kts	As potential warrants	60 minutes
Hail > 3/4”	As potential warrants	60 minutes
Lightning Within 5 NM	30 minutes	Issued When Lightning is Observed within 5 NM
Freezing Precipitation	As potential warrants	60 minutes
Heavy Snow > 2” in 12 hrs	As potential warrants	60 minutes
Wind Chill Temperature of -34F to -47F or colder	Observed advisories	Observed advisories
Wind Chill Temperature of -48F or colder	As potential warrants	Observed

Table A3.2. WWA MATRIX

WWA MATRIX			
	WWA	LT	ISSUED BY
WARNING	TORNADO ≤5SM	15	15 OWS
	SEVERE TS (≥50KT / ≥3/4" HAIL)	60	15 OWS
	MODERATE TS (≥35KT / ≥1/4" HAIL)	60	15 OWS
	WIND ≥50KT	60	15 OWS
	WIND ≥35KT BUT <50KT	60	15 OWS
	FREEZING PRECIP (ANY INTENSITY)	60	15 OWS
	BLIZZARD (3 HR DURATION, WND ≥30KT, VIS ≤1/4)	60	15 OWS
	HEAVY RAIN (>2" IN 12 HRS)	60	15 OWS
	HEAVY SNOW (>2" IN 12 HRS)	60	15 OWS
	LIGHTNING ≤5 SM	OBS	319 OSS/OSW
WATCH	TORNADO ≤5SM	POTENTIAL	15 OWS
	SEVERE TS (≥50KT / ≥3/4" HAIL)	POTENTIAL	15 OWS
	WIND ≥50KT	POTENTIAL	15 OWS
	FREEZING PRECIP (ANY INTENSITY)	POTENTIAL	15 OWS
	HEAVY RAIN (>2" IN 12 HRS)	POTENTIAL	15 OWS
	HEAVY SNOW (>2" IN 12 HRS)	POTENTIAL	15 OWS
	BLIZZARD (3 HR DURATION, WND ≥30KT, VIS ≤1/4)	POTENTIAL	15 OWS
	LIGHTNING ≤5 SM	POTENTIAL	15 OWS

Attachment 4

QUICK LOOK TEMPERATURE REACTION MATRIX

A4.1. This chart is meant as a "quick look" reference chart only; it is not a comprehensive chart. There may be more detailed instructions in the text or each group's appropriate attachment. It does not in any way relieve personnel from reading this document in its entirety.

Table A4.1. Wing Quick Look Temperature Reaction Matrix

TEMPERATURE *All temps are WCT*	**RG**	**MXG**	**MSG**	**MDG**	**ACTION**
+40º F			X		Child Development Center (CDC) restricts outdoor activity for children younger than 12 months.
+32º F	X	X	X	X	Do not wear cotton inserts inside Chemical Warfare Gloves (Attachment 13).
+20º F	X	X	X	X	**"MODERATE" COLD STRESS CONDITION** Reference Attachment 7 for illness prevention measures Visit unoccupied facilities per Attachment 14. Personnel, who work outside, follow procedures in Attachment 7. Initiate emergency use of NGH Heaters & warming shelters (Attachment 8).
0º F	X	X	X	X	Do not wear M50 gas mask outside (Attachment 13).
0º F			X		CDC restricts outdoor activity for all children CES performs all outside tasks w/ shelters available.
-15º F to -33º F	X				OSS/OSW issues wind chill advisory.
-20º F	X	X	X	X	**"CAUTION" COLD STRESS CONDITION** Reference Attachment 7 for illness prevention measures.
-20º F	X	X			ID and terminate non-essential outdoor activity Limit outside activities to aircraft identified in weekly schedule.
-20º F			X		ID and terminate non-essential outdoor activity. CES: terminate routine maintenance if no warming capability exists. LRS: vehicle recovery/mobile maintenance will continue w/ caution, verify road conditions and be aware of travel restrictions.
-30º F	X	X	X	X	Buddy System is in effect (Attachment 7).
-34º F to -47º F	X				OSS/OSW issues wind chill advisory.

-40º F	X				If needed coordination must be made with requesting agency, WF, and supporting OWS. Hangaring aircraft or substitution with aircraft already prepared for flight should be considered to generate essential sorties. See Attachment 2 for additional guidance.
-40º F		X		X	Outdoor ancillary training, aircraft maintenance qualification program courses and field training detachment courses canceled Priority Maintenance only.
-40º F			X	X	CES and LRS: Perform only mission tasks if warming capabilities are on site.
-40º F					LRS: Vehicle recovery/mobile maintenance response to direct mission-essential vehicles only. LRS: Verify road conditions and be aware of 319 RW travel restrictions. FSS: Terminate all delivery services.
-40º F	X	X	X	X	Accomplish unoccupied building checks every 8 hours or as required (Attachment 14). All normal outdoor training cancelled.
-48º F	X	X		X	OSS/OSW issues observed wind chill warning.
-50º F	X	X	X	X	**"DANGER" COLD STRESS CONDITION** Reference Attachment 7 for illness prevention measures.
-50º F	X	X		X	Outside maintenance restricted to national emergencies and peacetime contingencies only. Equipment and vehicles will not be driven except in an emergency, unless they are kept in warm garage, time outside is less than 1 hour, and vehicle is returned to warm garage. Approve emergency use and dispatch of AGE heaters off of the flight line to support all possible organizational emergencies.
-50º F				X	Will adjust clinic appointments as required and requested to ensure patient/staff safety. Outside activities are restricted to national emergencies and peace time contingencies only.
-50º F			X	X	LRS provide required TRNS per para 2.16.1.

Attachment 5

WIND CHILL TEMPERATURE (WCT) INDEX CHART

A5.1. WCTs are derived from calibrated weather equipment. We can reference chart for manual observing back-up procedures.

Table A5.1. WIND CHILL TEMPERATURE (WCT) INDEX CHART

Wind Speed (mph)	Temperature (°F)																	
Calm	**40**	**35**	**30**	**25**	**20**	**15**	**10**	**5**	**0**	**-5**	**-10**	**-15**	**-20**	**-25**	**-30**	**-35**	**-40**	**-45**
5	36	31	25	19	13	7	1	-5	-11	-16	-22	-28	-34	-40	-46	-52	-57	-63
10	34	27	21	15	9	3	-4	-10	-16	-22	-28	-35	-41	-47	-53	-59	-66	-72
15	32	25	19	13	6	0	-7	-13	-19	-26	-32	-39	-45	-51	-58	-64	-71	-77
20	30	24	17	11	4	-2	-9	-15	-22	-29	-35	-42	-48	-55	-61	-68	-74	-81
25	29	23	16	9	3	-4	-11	-17	-24	-31	-37	-44	-51	-58	-64	-71	-78	-84
30	28	22	15	8	1	-5	-12	-19	-26	-33	-39	-46	-53	-60	-67	-73	-80	-87
35	28	21	14	7	0	-7	-14	-21	-27	-34	-41	-48	-55	-62	-69	-76	-82	-89
40	27	20	13	6	-1	-8	-15	-22	-29	-36	-43	-50	-57	-64	-71	-78	-84	-91
45	26	19	12	5	-2	-9	-16	-23	-30	-37	-44	-51	-58	-65	-72	-79	-86	-93
50	26	19	12	4	-3	-10	-17	-24	-31	-38	-45	-52	-60	-67	-74	-81	-88	-95
55	25	18	11	4	-3	-11	-18	-25	-32	-39	-46	-54	-61	-68	-75	-82	-89	-97
60	25	17	10	3	-4	-11	-19	-26	-33	-40	-48	-55	-62	-69	-76	-84	-91	-98

Moderate: cold injuries (including frostbite) possible after long-term exposure	Caution: frostbite likely after **10-30 minutes** of exposure	Danger: frostbite likely after **5 minutes** of exposure

A5.2. This WCT chart is worst case approximations based on mathematical models and data derived from limited human studies. The exact effect of cold exposure on an individual will vary depending on the actual activity level, length of exposure, moderating effects of clothing or partial shelter from the wind and overall physical state.

A5.3. Freezing cold injury can occur anytime temperatures (air or surface) fall below freezing (32°F). However, the likelihood and severity of injury will increase with prolonged exposure to lower temperatures and greater air velocity, whether it is due to vehicle speed or actual wind.

A5.4. WCT is an indication of the combined effects of wind and colder air temperature. Individual level of activity, protective clothing, partial shelter from the wind, and overall fitness may alter the actual impact of cold exposure. Even when temperatures are above freezing, prolonged exposure to cold and wind may result in hypothermia or non-freezing cold injury (chilblains or trench foot).

Attachment 6

MINIMUM EXTREME COLD WEATHER PROTECTIVE ISSUE

A6.1. Commanders may issue additional items to this list in order to safely meet mission requirements and are responsible for determining their organization's temporary issue procedures. "Carhartt" style gear does not meet the intent of this instruction. The below listed items are also considered the minimum required for the Type B Bag, Extreme Cold Weather, for deployment purposes.

Table A6.1. Minimum Extreme Cold Weather Protective Issue

NOMENCLATURE	STOCK NUMBER	QT	UNIT	NOTE
Parka, APEC	8415-00-376-1XXX	1	EA	A,F
Parka, Level 7 (Gen 3)	8415-01-545-8718	1	EA	F, G
Trouser, Cold Weather	8415-00-394-3XXX	1	PR	A,G
Mitten Set	8415-00-782-671X	1	EA	A,B,H
Cap, Cold Weather	8415-01-099-784X	1	EA	A,C,H
Boots, Extreme Cold	8430-00-655-55XX	1	PR	A,H
Socks, Cold Weather	8440-00-153-67XX	2	PR	A,H
Kit Bag, Flyers (A-3)	8460-00-606-8366	1	EA	D

NOTE CODES:
These are sized items. Contact the Envision Store for specific NSNs and sizes.
A. Replaces NSN 8415-00-268-831X through attrition.
B. Replaces NSN 8415-00-782-29XX and NSN 8415-00-269-048X through attrition.
C. Replaces NSN 8465-00-261-6909 through attrition.
D. May be substituted with NSN 8415-01-228-13XX and SN 72065-990 (Parka, Gortex Cold Weather and Liner, Fleece (Top)
E. May be substituted with NSN 8415-01-228-13XX and SN 72420-990 (Trouser, Gortex Cold Weather and Liner, Fleece (Bottom)
F. May be substituted with equivalently rated item as determined by the organizational commander.
G. May consider for temporary issue to personnel.

Attachment 7

EXTREME COLD ILLNESS PREVENTION

A7.1. All Grand Forks AFB personnel will adhere to the procedures listed below for the prevention of cold stress condition illnesses while working outdoors.

Table A7.1. Extreme Cold Illness Prevention

Cold Stress Condition	Work/ Warming Cycle[2]	Required Water[3]	Work Procedures
"MODERATE" WCT[1]: +20º F to -19º F (Frostbite possible in 2 hours)	50 min work/ 20 min warm	1/2 quart per hour	- Cover metal handles with thermal insulation. Wear insulated gloves and cold weather clothing as required to avoid shivering and painful, cold skin. Avoid heavy sweating. - No outdoor operations involving water. Change wet clothes immediately. Maintain communication with all personnel, preferably through buddy system. - When wearing rubber gloves, do NOT wear cotton inserts; wear wool inserts instead. Wear cold weather clothing OVER ground crew ensemble as needed for protection from cold. Carry mask UNDER outer garments so it remains pliable. During exercises, remove mask and dry face before going outside, to prevent freezing perspiration. For exercises only, do NOT wear mask outdoors below 0º F, and do NOT wear rubber gloves below -10º F.
"CAUTION" WCT: -20º F to -49º F (Frostbite likely in < 45 minutes)	30 min work/ 20 min warm	¾ quart per hour	- Follow all "Moderate" precautions. Wear mittens over wool inserts, instead of gloves. Cover all exposed skin. Avoid wearing cotton garments. Wear wicking socks/glove liners. Change boot insoles daily or wear vapor barrier boots. Buddy system is mandated for all operations at-30° F and below.
"DANGER" WCT: < -50ºF (Frostbite in < 5 minutes)	10 min work/ 20 min warm	1 quart per hour	EMERGENCY KEY ACTIVITIES ONLY, as determined by RG/CC, GP/CC or ICC. Follow all precautions above.

NOTES:
(1)WCT is the wind chill temperature, obtained from your UCC, the CP, or the ICC.
(2)Warm breaks are to be taken in building, vehicles, or shelters heated to approximately 70ºF.
(3)Avoid caffeine (e.g., tea, coffee, soda) and alcohol. They dehydrate you. Instead, drink cool water and warm fluids such as broth, cider, or decaffeinated coffee, tea, or cocoa.

Attachment 8

USE OF NEW GENERATION HEATERS IN WARMING SHELTERS

A8.1. During +20º F temperatures and colder outdoor workers will take warm breaks in buildings, vehicles or shelters heated to about 70º F (a variance of no more than 10 degrees from this temperature is authorized).

A8.2. If a permanent, heated structure is not available nearby, supervisors must establish warming shelters. Contact 69 MXS AGE for information on using NGH to warm the shelters. Heaters should only be used for facilities during emergency conditions. NGH are authorized for flight line support only and emergency use. When using supplemental heaters, personnel will adhere to the following:

A8.2.1. Take proper precautions to ensure personnel are not overexposed to diesel exhaust. Personnel should exit the shelter any time they feel light-headed, dizzy, and/or nauseous. If symptoms persist or warrant medical attention, dial 911 immediately for prompt medical care.

A8.2.2. NGH must be placed downwind from the shelter, the heater intake facing into wind, and a minimum of two heating ducts (in series) used to avoid excessive temperatures.

A8.2.3. Users should not adjust NGH thermostats. Contact 69 MXS AGE to exchange NGH if they are not providing adequate heat.

Attachment 9

WINTER STORMS AND EXTREME COLD SAFETY

A9.1. Heavy snowfall and extreme cold can immobilize an entire region. Even areas that normally experience mild winters can be hit with a major snowstorm or extreme cold. Winter storms can result in flooding, storm surge, closed highways, blocked roads, downed power lines and hypothermia. This information will help you survive and operate at home or work during winter storms and extreme cold.

A9.2. Before Winter Storms and Extreme Cold

A9.2.1. Add the following items to your winter kit: rock salt to melt ice on walkways, sand to improve traction, snow shovels and other snow removal equipment.

A9.2.2. Prepare your home and family.

A9.2.2.1. Personnel should prepare for potential isolation in their home as a result of severe weather conditions. Personnel residing off base should plan for regular fuel sources being cut off in severe weather by having a sufficient back-up supply of heating fuel. For example, store a good supply of dry, seasoned wood for your fireplace or wood-burning stove.

A9.2.2.2. Winterize your home to extend the life of your fuel supply by insulating walls and attics, caulking and weather-stripping doors and windows, and installing storm windows or covering windows with plastic.

A9.2.2.3. Winterize your house, barn, shed or any other structure that may provide shelter for your family, neighbors, livestock or equipment. Clear rain gutters; repair roof leaks and cut away tree branches that could fall on a house or other structure during a storm.

A9.2.2.4. Insulate pipes with insulation or newspapers and plastic and allow faucets to drip a little during cold weather to avoid freezing.

A9.2.2.5. Keep fire extinguishers on hand, and make sure everyone in your house knows how to use them. House fires pose an additional risk, as more people turn to alternate heating sources without taking the necessary safety precautions.

A9.2.2.6. Learn how to shut off water valves (in case a pipe bursts).

A9.2.2.7. Know ahead of time what you should do to help elderly or disabled friends, neighbors, or employees.

A9.2.2.8. Hire a contractor to check the structural ability of the roof to sustain unusually heavy weight from the accumulation of snow - or water, if drains on flat roofs do not work.

A9.2.2.8.1. Personnel residing on base will comply with privatized housing requirements outlined by Balford Beatty Communities. If leaving the area during the winter months for more than 24 hours, personnel are required to obtain a “winter watchman” from the Base Housing Office. Personnel will be required to provide Balford Beatty Communities with the name of the person watching their house and taking care of the house’s exterior responsibilities.

A9.2.2.8.2. The winter watchman is a thermostat, red light bulb and extension cord kit. It is hung in a front window and it will let Balford Beatty maintenance technicians know when it gets too cold the house. If the light illuminates, maintenance will call whoever is watching the house to arrange a time to check the furnace operation and make sure there aren't any frozen water lines.

A9.2.3. Prepare your car by checking or having a mechanic check the following items on your car.

A9.2.3.1. Antifreeze levels - ensure they are sufficient to avoid freezing.

A9.2.3.2. Battery and ignition system - should be in top condition and battery terminals should be clean.

A9.2.3.3. Brakes - check for wear and fluid levels.

A9.2.3.4. Exhaust system - checks for leaks and crimped pipes and repair or replace as necessary. Carbon monoxide is deadly and usually gives no warning.

A9.2.3.5. Fuel and air filters - replace and keep water out of the system by using additives and maintaining a full tank of gas.

A9.2.3.6. Heater and defroster - ensure they work properly.

A9.2.3.7. Lights and flashing hazard lights - check for serviceability.

A9.2.3.8. Oil - check for level and weight. Heavier oils congeal more at low temperatures and do not lubricate as well.

A9.2.3.9. Thermostat - ensure it works properly.

A9.2.3.10. Windshield wiper equipment - repair any problems and maintain proper washer fluid level.

A9.2.3.11. Install good winter tires. Make sure the tires have adequate tread. All-weather radials are usually adequate for most winter conditions.

A9.2.3.12. Maintain at least a half tank of gas during the winter season.

A9.2.3.13. Place a winter emergency kit in each car that includes:

Table A9.1. Winter Emergency Kit

shovel	windshield scraper and small broom
flashlight	battery powered radio
extra batteries	water
snack food	matches
extra hats, socks and mittens	first aid kit with pocket knife
necessary medications	blanket(s)
tow chain or rope	road salt and sand
booster cables	emergency flares
fluorescent distress flag	

A9.2.4. Dress for the Weather

A9.2.4.1. Wear several layers of loose fitting, lightweight, warm clothing rather than one layer of heavy clothing. The outer garments should be tightly woven and water repellent.

A9.2.4.2. Wear mittens, which are warmer than gloves.

A9.2.4.3. Wear a hat.

A9.2.4.4. Cover your mouth with a scarf to protect your lungs.

A9.3. During a Winter Storm:

A9.3.1. Listen to your radio, television, or National Oceanic and Atmospheric Administration Weather Radio for weather reports and emergency information in addition to monitoring base severe weather information.

A9.3.2. Eat regularly and drink ample fluids, but avoid caffeine and alcohol.

A9.3.3. Conserve fuel, if necessary, by keeping your residence cooler than normal. Temporarily close off heat to some rooms.

A9.3.4. If the pipes freeze, remove any insulation or layers of newspapers and wrap pipes in rags. Completely open all faucets and pour hot water over the pipes, starting where they were most exposed to the cold (or where the cold was most likely to penetrate).

A9.3.5. Maintain ventilation when using kerosene heaters to avoid build-up of toxic fumes. Refuel kerosene heaters outside and keep them at least three feet from flammable objects.

A9.3.6. If you are outdoors:

A9.3.6.1. Avoid overexertion when shoveling snow. Overexertion can bring on a heart attack a major cause of death in the winter. If you must shovel snow, stretch before going outside.

A9.3.6.2. Cover your mouth. Protect your lungs from extremely cold air by covering your mouth when outdoors. Try not to speak unless absolutely necessary.

A9.3.6.3. Keep dry. Change wet clothing frequently to prevent a loss of body heat. Wet clothing loses all of its insulating value and transmits heat rapidly.

A9.3.6.4. Watch for signs of frostbite. These include loss of feeling and white or pale appearance in extremities such as fingers, toes, ear lobes, and the tip of the nose. If symptoms are detected, get medical help immediately.

A9.3.6.5. Watch for signs of hypothermia. These include uncontrollable shivering, memory loss, disorientation, incoherence, slurred speech, drowsiness, and apparent exhaustion.

A9.3.7. If you are driving

A9.3.7.1. Drive only if it is absolutely necessary.

A9.3.7.2. If you must drive, travel in the day, don’t travel alone, and keep others informed of your schedule. Stay on main roads; avoid back road shortcuts.

A9.3.8. If a blizzard traps you in the car:

A9.3.8.1. Pull off the highway. Turn on hazard lights and hang a distress flag from the radio antenna or window.

A9.3.8.2. Remain in your vehicle where rescuers are most likely to find you. Do not set out on foot unless you can see a building close by where you know you can take shelter. Be careful; distances are distorted by blowing snow. A building may seem close, but be too far to walk to in deep snow.

A9.3.8.3. Run the engine and heater about 10 minutes each hour to keep warm. When the engine is running, open a downwind window slightly for ventilation and periodically clear snow from the exhaust pipe. This will protect you from possible carbon monoxide poisoning.

A9.3.8.4. Exercise to maintain body heat, but avoid overexertion. In extreme cold, use road maps, seat covers, and floor mats for insulation. Huddle with passengers and use your coat for a blanket.

A9.3.8.5. Take turns sleeping. One person should be awake at all times to look for rescue crews.

A9.3.8.6. Drink plenty of fluids to avoid dehydration.

A9.3.8.7. Be careful not to waste battery power. Balance electrical energy needs (the use of lights, heat, and radio) with supply.

A9.3.8.8. Turn on the inside light at night so work crews or rescuers can see you.

A9.3.8.9. If stranded in a remote area, stomp large block letters in an open area spelling out HELP or SOS and line with rocks or tree limbs to attract the attention of rescue personnel who may be surveying the area by airplane.

A9.3.8.10. Leave the car and proceed on foot, if necessary, once the blizzard passes.

A9.4. The human body cannot adapt to cold except through proper clothing, shelter, and behavior changes.

A9.4.1. Cold weather can lower body temperature, resulting in impaired performance and cold injuries. Cold weather is often accompanied by wind, rain, snow, and ice, which can worsen the effects of cold. Inactivity for long periods increases the risk of cold injury. This is a particular concern for Defensive Fighting Positions (DFPs) or small vehicle crew compartments where movement is restricted. Minimize the risk of cold injuries in DFPs by placing pads, tree boughs, etc., inside these positions.

A9.4.1.1. For any given air temperature, the potential for body-heat loss, skin cooling, and decreased body temperature is increased by wind and moisture. Any source of nicotine, such as smoking or chewing tobacco, can increase susceptibility to cold injury. An early warning sign of inadequate clothing and shelter is shivering, which increases heat production. Over the counter drugs such as aspirin, ibuprofen, and acetaminophen, can interfere with shivering.

A9.4.1.2. Vigorous physical activity will also increase heat production, but adds to the risk of cold injury if clothes become wet with sweat. When body heat loss exceeds the body's ability to produce and retain heat, body temperature decreases. When body temperature falls below 95º F, hypothermia, a life threatening condition, follows. To reduce heat loss, the body decreases blood flow to the arms, legs, and skin. Although this

protects the internal organs, the decreased blood flow increases susceptibility of hands, feet, ears, etc., to nonfreezing (trench foot) and freezing (frostbite) cold injuries.

A9.4.2. Clean, dry, loose layers of clothing, including insulated headgear (watch caps/hoods), multiple gloves, and double socks, help prevent heat loss. The outer layer should provide wind and water protection. Adjust layers to minimize sweating. If clothing, including socks, becomes wet, change them immediately. Size boots to accommodate extra socks without restricting blood circulation. Cover exposed skin to help prevent frostbite, beginning at "moderate" cold stress levels.

A9.4.3. Metal objects and liquid fuels left in the cold can pose a serious hazard. Fuels and solvents remain liquid at very low temperatures. Skin contact with fuel or metal at below freezing temperatures can result in nearly instantaneous freezing injury.

A9.4.4. Maintain adequate food consumption to make up for increased energy requirements in cold weather. Eat normal meals with frequent nutritious snacks between meals. Maintain adequate hydration with fluid intake of 0.5 to 1 liter per hour. The low absolute humidity associated with adverse cold will cause dehydration with normal breathing. Avoid caffeine, tobacco, and alcohol. Keep hands, feet, and skin dry.

A9.5. Cold stress leads to reduced performance and lack of concentration. People are not always the best judge of their own condition. Supervisors must be aware of the risks of cold weather operations and enforce the buddy system to help prevent cold injuries. Individual tolerance to cold exposure varies greatly for a variety of reasons, including genetic and racial differences. Low body fat decreases tolerance, while higher body fat is protective. Use the recommended warming cycles in **Attachment 7** as well as the response of individuals to trigger actions to prevent cold injuries.

A9.6. Symptoms of Cold Injuries.

A9.6.1. Chilblain: A nonfreezing cold injury, which, although painful, causes little or no permanent impairment. It appears as tender, red, swollen skin that is hot to the touch and may itch. This can worsen to an aching, prickly ("pins and needles") sensation and then numbness. It may develop in only a few hours in skin exposed to cold.

A9.6.2. Immersion or Trench Foot: Develops when feet are exposed to moisture and cold for prolonged periods (12 hours or longer). The combination of cold and moisture softens skin, causing tissue loss and, often, infection. Untreated, trench foot can eventually require amputation. Often, the first sign of trench foot is itching, numbness or tingling pain. Later the feet may appear swollen, and the skin faintly red, blue, or black. Commonly, trench foot shows a distinct "waterline" coinciding with the water level in the boot.

A9.6.3. Frost Nip: Involves freezing of water on the skin surface. The skin will become reddened and possibly swollen. Although painful, there is usually no further damage after re-warming. Repeated frost nip can dry the skin, causing it to crack and be sensitive. Frost nip should be taken seriously since it may be the first sign of impending frostbite.

A9.6.4. Frostbite: Involves freezing of deeper layers of tissue. Ice crystal formation and lack of blood flow cause tissue damage. Skin freezes at about 28º F. The skin becomes numb and turns a gray or waxy white color, is cold to the touch, and may feel stiff.

A9.6.5. Hypothermia: An immediately life-threatening condition in which body temperature falls below 95º F. Generally, body temperature will not fall until after many hours of exposure to cold air or shorter exposure to cold water. Body temperature can fall even when air temperatures are above freezing if conditions are windy, clothing is wet, and/or the person is inactive. The first signs of developing hypothermia include confusion, bizarre behavior, and withdrawal from group interaction. Victims of hypothermia may be unconscious, with nearly undetectable breathing and pulse.

A9.7. First Aid for Cold Injuries

A9.7.1. Chilblain and Trench Foot: Prevent further cold exposure. Remove wet or constrictive clothing. Gently wash, dry and elevate the injured part. Cover the injured area with layers of loose warm clothing to allow re-warming. Pain and blisters may develop. Do not pop blisters, do not apply lotions or creams, do not massage, do not expose to extreme heat, and do not allow victim to walk on injury. Seek medical attention.

A9.7.2. Frostbite: Prevent further cold exposure and remove wet or constrictive clothing. Gradually re-warm the injury by direct skin-to-skin contact between injured area and non-injured skin of victim or a buddy. Evacuate for medical treatment. Victims with foot injuries should not walk, but should be evacuated by litter. Do not thaw frostbite injuries if there is a possibility of refreezing during evacuation.

A9.7.3. Hypothermia: Prevent further cold exposure and remove wet clothing. Initiate Cardio Pulmonary Resuscitation (CPR) if required. Transport for medical treatment as soon as possible. Re-warm by covering with blankets, sleeping bags or body-to-body contact if necessary. Get the victim off the ground. Handle gently during treatment and evacuation because rough handling of hypothermic victims can cause dangerous irregular heartbeats or seizures. Apparently lifeless victims are never pronounced dead, even by physicians, until they are “warm and dead.”

Attachment 10

LIGHTNING SAFETY

A10.1. In the United States, there are an estimated 25 million lightning flashes each year. During the past 30 years, lightning killed an average of 62 people per year. This ties the average of 62 deaths per year caused by tornadoes. Yet because lightning usually claims only one or two victims at a time and doesn't cause mass destruction of property, it is underrated as a risk. While documented lightning injuries in the United States average about 300 per year, undocumented injuries are likely much higher.

A10.2. Watch for developing thunderstorms. Thunderstorms are most likely to develop on spring or summer days but can occur year round. As the sun heats the air, pockets of warmer air start to rise and cumulus clouds form. Continued heating can cause these clouds to grow vertically into towering cumulus clouds, often the first sign of a developing thunderstorm.

A10.3. Lightning can strike as far as 10 miles from an area where it is raining. That's about the distance you can hear thunder. If you can hear thunder, you are within striking distance. Seek safe shelter immediately.

A10.4. Most lightning deaths and injuries occur in the summer. Where organized outdoor sports activities take place. Coaches, camp counselors and other adults must stop activities at the first roar of thunder to ensure everyone time to get in a large building or enclosed vehicle. Leaders of outdoors events should have a written plan that all staff are aware of and enforce.

A10.5. If outdoors, avoid water, high ground, open spaces, and metal objects including electric wires, fences, machinery, motors, power tools, etc. Unsafe places include underneath canopies, small picnic or rain shelters, or near trees. Where possible, find shelter in a substantial building or in a fully enclosed metal vehicle such as a car, truck or a van with the windows completely shut. If lightning is striking nearby when you are outside, you should crouch down, put feet together, place hands over ears to minimize hearing damage from thunder, and avoid proximity (minimum of 15 feet) to other people.

A10.6. If inside a building, stay off corded phones, computers, and other electrical equipment that put you in direct contact with electricity. Stay away from pools, indoor or outdoor, tubs, showers and other plumbing. Buy surge suppressors for key equipment. Install ground fault protectors on circuits near water or outdoors. When inside, wait 30 minutes after the last strike, before going out again.

A10.7. If a person is struck by lightning, call 911 and get medical care immediately. Cardiac arrest and irregularities, burns, and nerve damage are common in cases where people are struck by lightning. However, with proper treatment, including CPR if necessary, most victims survive a lightning strike. You are in no danger helping a lightning victim. The charge will not affect you.

A10.8. Lightning is dangerous. With common sense, you can greatly increase your safety and the safety of those you are with. At the first clap of thunder, go in a large building or fully enclosed vehicle and wait from a weather observer perspective lightning is no longer observed 15 minutes after last clap of thunder is heard.

Attachment 11

HIGH WIND SAFETY

A11.1. Extreme winds brought on by tornados, hurricanes, thunderstorms, and winter storms can cause serious damage to your buildings and property, putting you and your family at risk. Don't wait until it's too late. Wind-proofing your building is a smart way to prevent damage.

A11.2. During high winds, wind speeds fluctuate and winds may change directions. This puts enormous pressure on all parts of a building. Keeping your building well sealed on the outside will prevent wind from getting into your building and causing damage. In addition, debris carried by high winds can penetrate your building, causing damage and possible injury.

A11.3. The roof is one of the most vulnerable parts of your building. The connection between the roof and walls must be strong enough to resist the "uplift" effect of strong winds. Make sure the roof sheathing is properly installed.

A11.4. Windows should be protected to reduce the risk of damage or injury caused by flying debris and pressure changes.

A11.5. Check to be sure entry doors are secured with at least three hinges and a deadbolt security lock.

A11.6. Trim the trees around your building that could pose a threat by falling or hitting your building. Put away any furniture, grills, tools, and anything else that could become a flying missile.

A11.7. Roll-up doors (the larger, the more prone to damage) are also vulnerable to damage from high winds. Ensure they are installed correctly and securely.

A11.8. Tornados in North Dakota are not a significant threat.

Attachment 12

LARGE HAIL SAFETY

A12.1. "Large Hail" is greater than or equal to three-quarter inch in diameter. Large hail can cause serious or even fatal injuries.

A12.2. If you are in an automobile, stop driving. If you can see a safe place close by (like inside a garage, under a highway overpass, or under a service station awning), drive there as soon as you can. Make sure you pull completely off the highway. Do not leave the vehicle until it stops hailing. Your car will furnish reasonable protection.

A12.3. Stay away from car windows. Cover your eyes with something (like a piece of clothing). If possible, get onto the floor face down, or lie down on the seat with your back to the windows. Put very small children under you, and cover their eyes.

A12.4. Stay inside until the hail stops if you are in a building. Stay away from the windows, especially those being struck by hail. Account for all family members, building occupants, pets, etc. Do not go outside for any reason. Avoid using phones and electrical appliances during a severe storm to avoid the danger of electrocution from lightning.

A12.5. If you are caught outdoors, seek shelter immediately. If you can't find something to protect your entire body, at least find something to protect your head. Stay out of culverts and lowland areas that might fill suddenly with water. Trees are a last resort. It is common during severe storms for trees to lose branches. Also, large isolated trees attract lightning.

Attachment 13

CHEMICAL BIOLOGICAL WARFARE DEFENSE (CBWD) COLD WEATHER OPERATIONS

A13.1. As a Chemical Biological Radiological Nuclear and Explosive (CBRNE) Low Threat Area, Grand Forks AFB is not under immediate chemical or biological threat posed by nation states or non-state actors. The following guidelines, however, are to be used in training and exercising scenarios in which personnel may be required to utilize Personal Protective Equipment (PPE).

A13.2. Commanders should not consider cold conditions a deterrent to Chemical Warfare agent use. Agent physical properties are affected by temperature, but the effectiveness is not always degraded at very low temperatures. The major threat from the employment of chemical warfare weapons in cold regions is the delayed action. The "Danger" is created when nerve and blister agents are carried into heated shelters on clothing, footgear, or other equipment. The warmth of the shelter will reactivate these agents creating a chemical vapor and/or a liquid hazard in the enclosed area.

A13.3. Cold weather clothing (parka, field jacket, mittens, etc.) is to be worn OVER the ground crew ensemble (GCE) if the individual needs protection from the cold. This is an appropriate practice during exercises and war. Items worn over the GCE will become contaminated upon contact with CBRNE agents and must be properly discarded during decontamination procedures.

A13.4. Do not wear cotton glove inserts below 32° F. For temperatures between 32° F and - 10° F, personnel should wear polypropylene gloves with wool inserts under the rubber gloves. Personnel may also wear winter or leather gloves over the rubber gloves. When temperatures fall below -10° F, personnel should wear polypropylene gloves, wool inserts, and winter gloves (minus the rubber gloves).

A13.5. In extremely cold temperatures, the M50 protective masks become very rigid and are difficult to don. When donned at these temperatures, the protective mask causes instantaneous frostbite on the individual's face. Due to this potential, 319 RW personnel will not wear protective masks outdoors, during exercises, when wind chill temperatures (WCT) are 0º F or below. If moving from indoors to outdoors, remove mask and dry face prior to going outside. Personnel will wear the mask when indoors.

A13.6. When temperatures are above 0° F WCT, personnel will observe the appropriate Mission Oriented Protection Posture (MOPP) level, including wear of the gas mask. Below 32º F, personnel should be aware the gas mask may still be very cold and be somewhat rigid. Also, the mask eye lenses may fog and freeze when the mask is donned. In addition, breath condensation may freeze the valves and the voicemitter assembly. To minimize this, during MOPP conditions 0-2, the mask must be carried under outer garments to remain relatively warm from the individual's body heat.

Attachment 14

COLD WEATHER FACILITY INSPECTION PROCEDURES

A14.1. With the arrival of extended cold weather, it is essential to inspect unoccupied facilities to ensure there is adequate heat to prevent property loss from frozen water lines or leaking steam lines.

A14.2. Visit unoccupied facilities daily when temperatures are +20º F to -39º F. If temperatures are -40º F or colder, conduct building checks a minimum of every 8 hours. Inspect each unoccupied facility to ensure:

A14.2.1. Windows and exterior doors are completely closed and secure, with no excessive build-up of frost or ice on the inside. If signs of excessive frost build-up are present, look for sources of moisture like open water lines or steam leaks.

A14.2.2. Administrative and living space temperatures are minimum 60º F.

A14.2.3. Warehouse/hangar space temperatures are minimum 50º F.

A14.2.4. Temperature and humidity are under control. Temperatures greater than 80º F, high humidity, or musty odors may indicate a problem with the heating system, pumps, etc.

A14.2.5. Furnaces, blowers, pumps, etc. are not making unusual or loud noises.

A14.2.6. Interior doors are left open to maximize air movement within the facility.

A14.2.7. Check floors for standing water from frozen pipes. If there are no obvious signs of burst pipes, open faucets to ensure water flow.

A14.2.8. Inspect for steam leaks. Look around pipe fittings and valves for fluids.

A14.3. Facilities that do not have water and/or steam heat are exempt from the temperature-driven checks.

A14.4. If there are any problems, contact the 319 CES Customer Service desk.